**MTh # 401**

**Assignment no # 2**

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**Question;**

**:$(x^2+4)y''+y$ Solve the following differential equation by using power series method,**

**Solution**

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*Solve the following differential equation by using power series method*

$$\left(x^2+4\right)y''+y=0$$

sin *gular matrix is*

$$Q(x)=x^2+4$$

$$x=\pm 2$$

*according to power series*

$$y=\sum_{n=0}^{\infty}C_n x^n$$

$$y'=\sum_{n=1}^{\infty}nC_n x^{n-1}$$

$$y''=\sum_{n=2}^{\infty}n(n-1)C_n x^{n-2}$$

*putting the value in eq* $(1)$

$$\left(x^2+4\right)\sum_{n=2}^{\infty}n(n-1)C_n x^{n-2}+\sum_{n=0}^{\infty}C_n x^n=0$$

$$x^2\sum_{n=2}^{\infty}n(n-1)C_n x^{n-2}+4\sum_{n=2}^{\infty}n(n-1)C_n x^{n-2}+\sum_{n=0}^{\infty}C_n x^n=0$$

$$x^2x^{-2}\sum_{n=2}^{\infty}n(n-1)C_n x^{n}+4\sum_{n=2}^{\infty}n(n-1)C_n x^{n-2}+\sum_{n=0}^{\infty}C_n x^n=0$$

$$\sum_{n=2}^{\infty}n(n-1)C_n x^{n}+4\sum_{n=2}^{\infty}n(n-1)C_n x^{n-2}+\sum_{n=0}^{\infty}C_n x^n=0$$

exp *end by* $\sum_{n=2}^{\infty}n(n-1)C_n x^{n-2}$

$$4(2)(2-1)c_2x^{2-2}=8c_2x^0$$

$$4(2)(2-1)c_2x^{2-2}=8c_2x^0$$

$$4(3)(3-1)c_3x^{3-2}=24c_3x$$

$$y=\sum_{n=0}^{\infty}C_n x^n$$

$c_0x^0$ *and* $c_1x \Rightarrow \sum_{n=2}^{\infty}C_n x^n$

$$\sum_{n=2}^{\infty}n(n-1)C_n x^{n-2}+8c_2x^0+24c_3x+c_0x^0+c_1x+4\sum_{n=2}^{\infty}n(n-1)C_n x^{n-2}+\sum_{n=2}^{\infty}C_n x^n=0$$

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$$8c_2x^0+24c_3x+c_0x^0+c_1x+\sum_{n=2}^{\infty}n(n-1)c_nx_n+4\sum_{n=4}^{\infty}n(n-1)c_nx^{n-2}+\sum_{n=2}^{\infty}c_nx^n=0$$

*let* $n=k \quad n-2=k \quad n=k+2$

$$8c_2x^0+24c_3x+c_0x^0+c_1x+\sum_{k=2}^{\infty}k(k-1)c_kx_k+4\sum_{n=4}^{\infty}(k+2)(K+1)c_{k+2}x^k+\sum_{k=2}^{\infty}c_kx^k=0$$

$$8c_2x^0+24c_3x+c_0x^0+c_1x=0$$

$$\sum_{k=2}^{\infty}[k(k-1)c_k+4(k+2)(K+1)c_{k+2}+c_k]x^k=0$$

$$k(k-1)c_k+4(k+2)(k+1)c_{k+2}+c_k=0$$

$$c_{k+2}=\frac{-c_k-k(k-1)c_k}{4(k+2)(k+1)} \qquad k=1,2,3,4.....$$

$$c_2=\frac{-c_0-0(0-1)c_0}{4(0+2)(0+1)} \quad \Rightarrow \frac{-c_0}{4.2.1} \quad \Rightarrow \frac{-c_0}{2^2.2!}$$

$$c_3=\frac{-c_1-0(0-1)c_0}{4(0+2)(0+1)} \quad \Rightarrow \frac{-c_0}{4.2.1} \quad \Rightarrow \frac{-c_1}{2^2.3!}$$

$$c_4=\frac{-c_1-2(2-1)c_2}{4(2+2)(2+1)} \quad \Rightarrow \frac{-c_2-2c_2}{4.4.3} \quad \Rightarrow \frac{-3c_2}{2^2.4!} \quad \Rightarrow \frac{-3\left(\frac{-c_0}{2^2.2!}\right)}{2^4.4!} \quad =\frac{3c_0}{2^4.4!}$$

$$c_5=\frac{-c_3-3(3-1)c_3}{4(3+2)(3+1)} \quad \Rightarrow \frac{-c_3-6c_3}{4.4.3} \quad \Rightarrow \frac{-7c_3}{2^2.5!} \quad \Rightarrow \frac{-7\left(\frac{-c_1}{2^2.3!}\right)}{2^4.5!} \quad =\frac{7c_1}{2^4.5!}$$

$$c_6=\frac{-39}{2^66!}x^6$$

$$y=c_0+c_1x+c_1x^2+c_3x^3+c_4x^4+c_5x^5.....$$

*putting the value of in power series eq*

$$y=c_0+c_1x+\frac{-c_0}{2^2.2!}x^2-\frac{c_1}{2^2.3!}x^3+\frac{3c_0}{2^4.4!}x^4+\frac{7c_1}{2^4.5!}x^5-\frac{39}{2^66!}x^6.....$$

$$y=c_0+c_1x-\frac{c_0}{2^2.2!}x^2+\frac{-c_1}{2^2.3!}x^3+\frac{3c_0}{2^4.4!}x^4+\frac{7c_1}{2^4.5!}x^5-\frac{39}{2^66!}x^6.....$$

$$y=[c_0-\frac{c_0}{2^2.2!}x^2+\frac{7c_1}{2^4.4!}x^5-\frac{39}{2^66!}x^6.....]+[c_1x-\frac{3c_1}{2^2.3!}x^3+\frac{7c_1}{2^45!}x^5.....]$$

$$y=c_0[1-\frac{1}{2^2.2!}x^2+\frac{7}{2^4.4!}x^5-\frac{39}{2^66!}x^6.....]+c_1[x-\frac{3}{2^2.3!}x^3+\frac{7}{2^45!}x^5.....]$$

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